```
from huggingface_hub import login
login()
```

```
import os
import pandas as pd
import random
import matplotlib.pyplot as plt
import tensorflow as tf
import string
import re
import numpy as np
import nltk
from nltk.translate.bleu_score import sentence_bleu, SmoothingFunction
from nltk.translate.meteor_score import meteor_score
import seaborn as sns
import json
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

# Download NLTK data
nltk.download('punkt', quiet=True)
nltk.download('wordnet', quiet=True)
nltk.download('omw-1.4', quiet=True)

print(f"TensorFlow Version: {tf.__version__}")
gpu = len(tf.config.list_physical_devices('GPU')) > 0
print("GPU is", "AVAILABLE" if gpu else "NOT AVAILABLE")
```

```
TensorFlow Version: 2.19.0
GPU is AVAILABLE
```

```
# PART 1:  PREPROCESSING


class DataPreprocessor:
    def __init__(self):
        self.preprocessing_log = []
        self.stats = {}

    def log_step(self, step_name, details):
        self.preprocessing_log.append({
            'step': step_name,
            'timestamp': datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
            'details': details
        })
        print(f"  \u2713 {step_name}")

    def clean_text(self, text, text_type="source"):
        text = str(text)
        text = re.sub(r'<.*?>', '', text)
        text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
        text = re.sub(r'\S+@\S+', '', text)

        emoji_pattern = re.compile("["
            u"\U0001F600-\U0001F64F"
            u"\U0001F300-\U0001F5FF"
            u"\U0001F680-\U0001F6FF"
            u"\U0001F1E0-\U0001F1FF"
            u"\U00002702-\U000027B0"
            u"\U000024C2-\U0001F251"
            "]+", flags=re.UNICODE)
        text = emoji_pattern.sub(r'', text)

        contractions = {
            r"won't": "will not", r"can't": "cannot", r"n't": " not",
            r"'re": " are", r"'s": " is", r"'d": " would",
            r"'ll": " will", r"'t": " not", r"'ve": " have", r"'m": " am"
        }
        for pattern, replacement in contractions.items():
            text = re.sub(pattern, replacement, text, flags=re.IGNORECASE)

        text = re.sub(r'\s+', ' ', text).strip()
        return text

    def normalize_text(self, text):
        import unicodedata
        text = text.lower()
        text = re.sub(r'\d+', '<NUM>', text)
        text = unicodedata.normalize('NFKD', text).encode('ASCII', 'ignore').decode('ASCII')
```

```
        text = unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').decode('ascii')
        text = re.sub(r[""]', '"', text)
        text = re.sub(r"['\u2019]", "'", text)
        text = re.sub(r'[^\w\s<>"\']', ' ', text)
        text = re.sub(r'\s+', ' ', text).strip()
        return text

    def analyze_dataset(self, df, source_col, target_col):

        print("DATASET ANALYSIS")


        analysis = {
            'total_samples': len(df),
            'source_stats': {
                'avg_length': df[source_col].str.len().mean(),
                'max_length': df[source_col].str.len().max(),
                'min_length': df[source_col].str.len().min(),
                'unique_words': len(set(' '.join(df[source_col]).split())),
            },
            'target_stats': {
                'avg_length': df[target_col].str.len().mean(),
                'max_length': df[target_col].str.len().max(),
                'min_length': df[target_col].str.len().min(),
                'unique_words': len(set(' '.join(df[target_col]).split())),
            },
            'missing_values': {
                'source': df[source_col].isnull().sum(),
                'target': df[target_col].isnull().sum()
            },
            'duplicates': df.duplicated().sum()
        }

        print(f"  Total samples: {analysis['total_samples']}")
        print(f"  Duplicates: {analysis['duplicates']}")
        print(f"  Source avg length: {analysis['source_stats']['avg_length']:.1f} chars")
        print(f"  Target avg length: {analysis['target_stats']['avg_length']:.1f} chars")
        return analysis

    def prepare_data(self, df, source_col, target_col):
        print("PREPROCESSING PIPELINE")


        initial_analysis = self.analyze_dataset(df, source_col, target_col)
        self.stats['initial'] = initial_analysis

        print("\n Step 1: Handling missing values...")
        missing_source = df[source_col].isnull().sum()
        missing_target = df[target_col].isnull().sum()
        df = df.dropna(subset=[source_col, target_col])
        self.log_step('missing_value_handling', {
            'missing_source_removed': int(missing_source),
            'missing_target_removed': int(missing_target)
        })

        print("\n Step 2: Removing duplicates...")
        duplicate_count = df.duplicated().sum()
        df = df.drop_duplicates()
        self.log_step('duplicate_removal', {'duplicates_removed': int(duplicate_count)})

        print("\n Step 3: Cleaning texts...")
        df = df.copy()
        df['source_cleaned'] = df[source_col].apply(lambda x: self.clean_text(x, "source"))
        df['target_cleaned'] = df[target_col].apply(lambda x: self.clean_text(x, "target"))
        self.log_step('text_cleaning', {'samples': len(df)})

        print("\n Step 4: Normalizing texts...")
        df['source_normalized'] = df['source_cleaned'].apply(self.normalize_text)
        df['target_normalized'] = df['target_cleaned'].apply(self.normalize_text)
        self.log_step('text_normalization', {})

        print("\n Step 5: Filtering empty strings...")
        empty_source = (df['source_normalized'] == '').sum()
        empty_target = (df['target_normalized'] == '').sum()
        df = df[df['source_normalized'] != '']
        df = df[df['target_normalized'] != '']
        self.log_step('empty_filter', {
            'empty_source_removed': int(empty_source),
            'empty_target_removed': int(empty_target)
        })

        print("\n Step 6: Length-based filtering...")
        df = df[df['source_normalized'].str.len() <= 500]
        df = df[df['target_normalized'].str.len() <= 500]
```

```
        df = df[df['target_normalized'].str.len() <= 500]
        self.log_step('length_filtering', {'max_length_allowed': 500})

        final_analysis = self.analyze_dataset(df, 'source_normalized', 'target_normalized')
        self.stats['final'] = final_analysis

        documentation = {
            'timestamp': datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
            'initial_stats': initial_analysis,
            'final_stats': final_analysis,
            'preprocessing_steps': self.preprocessing_log,
            'data_reduction': {
                'initial_samples': initial_analysis['total_samples'],
                'final_samples': final_analysis['total_samples'],
                'reduction_percentage': (1 - final_analysis['total_samples'] / initial_analysis['total_samples'])
            }
        }

        def default_json_encoder(obj):
            if isinstance(obj, np.integer): return int(obj)
            elif isinstance(obj, np.floating): return float(obj)
            raise TypeError(f"Object of type {obj.__class__.__name__} is not JSON serializable")

        with open('preprocessing_documentation.json', 'w') as f:
            json.dump(documentation, f, indent=2, default=default_json_encoder)

        print(f"\n  Initial: {initial_analysis['total_samples']} -> Final: {final_analysis['total_samples']} samp
        print(f"  Documentation saved to 'preprocessing_documentation.json'")
        return df


print("# PART 1: DATA PREPROCESSING")

df = pd.read_csv('/content/data.csv', usecols=[1, 2])
df.columns = ['source', 'target']
df = df.astype(str)

preprocessor = DataPreprocessor()
df = preprocessor.prepare_data(df, 'source', 'target')
df['source'] = df['source_normalized']
df['target'] = df['target_normalized']

print("\nRandom samples after preprocessing:")
print(df[['source', 'target']].sample(5))
```

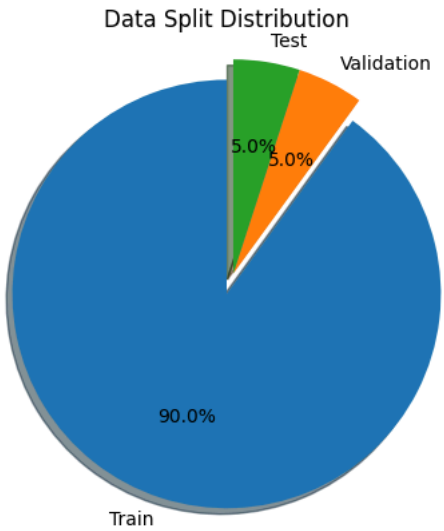
```
# PART 1: DATA PREPROCESSING
PREPROCESSING PIPELINE
DATASET ANALYSIS
  Total samples: 18587
  Duplicates: 2626
  Source avg length: 95.3 chars
  Target avg length: 264.1 chars

 Step 1: Handling missing values...
  ✓ missing_value_handling

 Step 2: Removing duplicates...
  ✓ duplicate_removal

 Step 3: Cleaning texts...
  ✓ text_cleaning

 Step 4: Normalizing texts...
  ✓ text_normalization

 Step 5: Filtering empty strings...
  ✓ empty_filter

 Step 6: Length-based filtering...
  ✓ length_filtering
DATASET ANALYSIS
  Total samples: 15073
  Duplicates: 0
  Source avg length: 92.5 chars
  Target avg length: 236.7 chars

  Initial: 18587 -> Final: 15073 samples
  Documentation saved to 'preprocessing_documentation.json'

Random samples after preprocessing:
                                                  source  \
7768   how can family members and caregivers support ...
6179   can nicotine interventions be considered in in...
3824   should individuals with a history of blood can...
```

```
13784  can environmental modifications help reduce ep...
8695   are there differences in alzheimer is risk fol...

                                                  target
7768   providing a calm and structured environment ma...
6179   considering nicotine interventions in individu...
3824   consult with an oncologist to determine person...
13784  environmental modifications such as removing p...
8695   while research is ongoing some studies suggest...
```

```
# PART 2: EXPERIMENT TRACKING

class ExperimentTracker:
    def __init__(self):
        self.experiments = []
        self.best_score = 0
        self.best_config = None
        self.baseline_performance = None

    def set_baseline(self, metrics):
        self.baseline_performance = metrics
        print(f"\n  Baseline - Accuracy: {metrics.get('accuracy', 0):.4f}  Loss: {metrics.get('loss', 0):.4f}")

    def add_experiment(self, config, metrics, notes=""):
        improvement = 0
        if self.baseline_performance:
            baseline_acc = self.baseline_performance.get('accuracy', 0)
            current_acc = metrics.get('val_accuracy', 0)
            if baseline_acc > 0:
                improvement = ((current_acc - baseline_acc) / baseline_acc) * 100

        experiment = {
            'timestamp': datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
            'config': config,
            'metrics': metrics,
            'improvement_over_baseline': f"{improvement:.2f}%",
            'notes': notes
        }
        self.experiments.append(experiment)

        if metrics.get('val_accuracy', 0) > self.best_score:
            self.best_score = metrics['val_accuracy']
            self.best_config = config
            print(f"  New best model! Val Accuracy: {self.best_score:.4f}")

        self.save_experiments()

    def save_experiments(self):
        with open('hyperparameter_experiments.json', 'w') as f:
            json.dump(self.experiments, f, indent=2)

    def display_comparison_table(self):
        print("\n" + "="*100)
        print("HYPERPARAMETER EXPERIMENT COMPARISON")
        print("="*100)

        comparison_data = []
        for exp in self.experiments:
            comparison_data.append({
                'Experiment': len(comparison_data) + 1,
                'Name': exp['config'].get('name', 'N/A'),
                'Embed': exp['config'].get('embed_dim', 'N/A'),
                'Dense': exp['config'].get('dense_dim', 'N/A'),
                'Heads': exp['config'].get('num_heads', 'N/A'),
                'Batch': exp['config'].get('batch_size', 'N/A'),
                'LR': exp['config'].get('learning_rate', 'N/A'),
                'Dropout': exp['config'].get('dropout', 'N/A'),
                'Val Acc': f"{exp['metrics'].get('val_accuracy', 0):.4f}",
                'Val Loss': f"{exp['metrics'].get('val_loss', 0):.4f}",
                'Improvement': exp['improvement_over_baseline'],
                'Status': exp['metrics'].get('status', 'OK')
            })

        comparison_df = pd.DataFrame(comparison_data)
        print(comparison_df.to_string(index=False))

        valid = [e for e in self.experiments if e['metrics'].get('status', 'OK') == 'OK']
        if valid:
            best_idx = np.argmax([e['metrics'].get('val_accuracy', 0) for e in valid])
            print(f"\n  Best: {valid[best_idx]['config']['name']}  "
                  f"Val Acc: {valid[best_idx]['metrics']['val_accuracy']:.4f}")
        return comparison_df


tracker = ExperimentTracker()
```

```
# VECTORIZATION

max_tokens = 25000
sequence_length = 200

strip_chars = string.punctuation.replace("[", "").replace("]", "")

def custom_standardization(input_string):
    lowercase = tf.strings.lower(input_string)
    return tf.strings.regex_replace(lowercase, "[" + re.escape(strip_chars) + "]", "")

source_vectorization = tf.keras.layers.TextVectorization(
    max_tokens=max_tokens,
    output_mode="int",
    output_sequence_length=sequence_length,
)

target_vectorization = tf.keras.layers.TextVectorization(
    max_tokens=max_tokens,
    output_mode="int",
    output_sequence_length=sequence_length + 1,
    standardize=custom_standardization,
)

# Shuffle and split
df = df.sample(frac=1).reset_index(drop=True)
train_size = int(len(df) * 0.90)
val_size   = int(len(df) * 0.05)

train_df = df[:train_size]
val_df   = df[train_size:train_size + val_size]
test_df  = df[train_size + val_size:]

print(f"\nSplit sizes -> Train: {len(train_df)}  Val: {len(val_df)}  Test: {len(test_df)}")

# Pie chart
labels = ['Train', 'Validation', 'Test']
sizes  = [len(train_df), len(val_df), len(test_df)]
fig1, ax1 = plt.subplots()
ax1.pie(sizes, explode=(0.1, 0, 0), labels=labels, autopct='%1.1f%%', shadow=True, startangle=90)
ax1.axis('equal')
plt.title('Data Split Distribution')
plt.show()

# Prepare text arrays
train_source_texts = train_df['source'].values
train_target_texts = ["[start] " + t + " [end]" for t in train_df['target'].values]

val_source_texts = val_df['source'].values
val_target_texts = ["[start] " + t + " [end]" for t in val_df['target'].values]

test_source_texts = test_df['source'].values
test_target_texts = ["[start] " + t + " [end]" for t in test_df['target'].values]

# Adapt on training data only
source_vectorization.adapt(train_source_texts)
target_vectorization.adapt(train_target_texts)
```

```
Split sizes -> Train: 13565  Val: 753  Test: 755
```



```
def make_dataset(source_texts, target_texts, batch_size):
    """
    Build a tf.data pipeline.
    FIX: cache() BEFORE shuffle() so the dataset is cached in its original
    order and then shuffled each epoch — avoids freezing the shuffle.
    """
    src = source_vectorization(source_texts)
    tgt = target_vectorization(target_texts)

    enc_input = src
    dec_input = tgt[:, :-1]
    dec_target = tgt[:, 1:]

    dataset = tf.data.Dataset.from_tensor_slices((
        {"source": enc_input, "target": dec_input},
        dec_target
    ))
    return (
        dataset
        .cache()                                # cache before shuffle
        .shuffle(buffer_size=len(source_texts))
        .batch(batch_size, drop_remainder=False)
        .prefetch(tf.data.AUTOTUNE)
    )
```

```
# TRANSFORMER COMPONENTS
class PositionalEmbedding(tf.keras.layers.Layer):
    def __init__(self, sequence_length, input_dim, output_dim, **kwargs):
        super().__init__(**kwargs)
        self.token_embeddings = tf.keras.layers.Embedding(
            input_dim=input_dim, output_dim=output_dim, mask_zero=True)
        self.position_embeddings = tf.keras.layers.Embedding(
            input_dim=sequence_length, output_dim=output_dim)
        self.sequence_length = sequence_length
        self.input_dim = input_dim
        self.output_dim = output_dim

    def call(self, inputs):
        length = tf.shape(inputs)[-1]
        positions = tf.range(start=0, limit=length, delta=1)
        return self.token_embeddings(inputs) + self.position_embeddings(positions)

    def compute_mask(self, inputs, mask=None):
        return self.token_embeddings.compute_mask(inputs)

    def get_config(self):
        config = super().get_config()
        config.update({"input_dim": self.input_dim, "output_dim": self.output_dim,
                       "sequence_length": self.sequence_length})
        return config
```

```
def shape_list(x):
    static = x.shape.as_list()
    dynamic = tf.shape(x)
    return [dynamic[i] if s is None else s for i, s in enumerate(static)]


def attention_mask(nd, ns, *, dtype):
    i = tf.range(nd)[:, None]
    j = tf.range(ns)
    return tf.cast(i >= j - ns + nd, dtype)


def mask_attn_weights(w):
    _, _, nd, ns = shape_list(w)
    b = tf.reshape(attention_mask(nd, ns, dtype=w.dtype), [1, 1, nd, ns])
    return w * b - tf.cast(1e10, w.dtype) * (1 - b)


def scaled_dot_product_attention(q, k, v, use_causal_mask=False, key_padding_mask=None):
    d_k = tf.cast(tf.shape(k)[-1], tf.float32)
    scaled_scores = tf.matmul(q, k, transpose_b=True) / tf.math.sqrt(d_k)
    if use_causal_mask:
        scaled_scores = mask_attn_weights(scaled_scores)
    if key_padding_mask is not None:
        key_padding_mask = tf.cast(key_padding_mask[:, tf.newaxis, tf.newaxis, :], scaled_scores.dtype)
        scaled_scores += (key_padding_mask - 1) * 1e9
    return tf.matmul(tf.nn.softmax(scaled_scores, axis=-1), v)
```

```
class MultiHeadAttention(tf.keras.layers.Layer):
    def __init__(self, embed_dim, h, **kwargs):
        super().__init__(**kwargs)
        self.embed_dim = embed_dim
        self.h = h
        if embed_dim % h != 0:
            raise ValueError(f"embed_dim ({embed_dim}) must be divisible by num_heads ({h})")
        self.q_linear = tf.keras.layers.Dense(embed_dim)
        self.k_linear = tf.keras.layers.Dense(embed_dim)
        self.v_linear = tf.keras.layers.Dense(embed_dim)
        self.concat_linear = tf.keras.layers.Dense(embed_dim)

    def split_heads(self, x, batch_size):
        x = tf.reshape(x, (batch_size, -1, self.h, self.embed_dim // self.h))
        return tf.transpose(x, perm=[0, 2, 1, 3])

    def concat_heads(self, x, batch_size):
        x = tf.transpose(x, perm=[0, 2, 1, 3])
        return tf.reshape(x, (batch_size, -1, self.embed_dim))

    def call(self, q, k, v, use_causal_mask=False, key_padding_mask=None):
        batch_size = tf.shape(k)[0]
        q = self.split_heads(self.q_linear(q), batch_size)
        k = self.split_heads(self.k_linear(k), batch_size)
        v = self.split_heads(self.v_linear(v), batch_size)
        attention = scaled_dot_product_attention(q, k, v, use_causal_mask, key_padding_mask)
        return self.concat_linear(self.concat_heads(attention, batch_size))

    def get_config(self):
        config = super().get_config()
        config.update({"embed_dim": self.embed_dim, "h": self.h})
        return config
```

```
class TransformerEncoder(tf.keras.layers.Layer):
    def __init__(self, embed_dim, dense_dim, num_heads, **kwargs):
        super().__init__(**kwargs)
        self.embed_dim = embed_dim
        self.dense_dim = dense_dim
        self.num_heads = num_heads
        self.layer_norm_1 = tf.keras.layers.LayerNormalization()
        self.layer_norm_2 = tf.keras.layers.LayerNormalization()
        self.global_self_attention = MultiHeadAttention(embed_dim=embed_dim, h=num_heads)
        self.feed_forward = tf.keras.Sequential([
            tf.keras.layers.Dense(dense_dim, activation="relu"),
            tf.keras.layers.Dense(embed_dim),
        ])
        self.supports_masking = True

    def call(self, x, mask=None):
        x = self.layer_norm_1(x + self.global_self_attention(q=x, k=x, v=x, key_padding_mask=mask))
        x = self.layer_norm_2(x + self.feed_forward(x))
        return x

    def get_config(self):
        config = super().get_config()
        config.update({"embed_dim": self.embed_dim, "dense_dim": self.dense_dim, "num_heads": self.num_heads})
        return config
```

```
class TransformerDecoder(tf.keras.layers.Layer):
    def __init__(self, embed_dim, dense_dim, num_heads, **kwargs):
        super().__init__(**kwargs)
        self.embed_dim = embed_dim
        self.dense_dim = dense_dim
        self.num_heads = num_heads
        self.causal_self_attention = MultiHeadAttention(embed_dim=embed_dim, h=num_heads)
        self.cross_attention = MultiHeadAttention(embed_dim=embed_dim, h=num_heads)
        self.feed_forward = tf.keras.Sequential([
            tf.keras.layers.Dense(dense_dim, activation="relu"),
            tf.keras.layers.Dense(embed_dim),
        ])
        self.layer_norm_1 = tf.keras.layers.LayerNormalization()
        self.layer_norm_2 = tf.keras.layers.LayerNormalization()
        self.layer_norm_3 = tf.keras.layers.LayerNormalization()
        self.supports_masking = True

    def call(self, x, context, mask=None, context_mask=None):
        x = self.layer_norm_1(
            x + self.causal_self_attention(q=x, k=x, v=x, use_causal_mask=True, key_padding_mask=mask))
        x = self.layer_norm_2(x + self.cross_attention(q=x, k=context, v=context, key_padding_mask=context_mask)
        x = self.layer_norm_3(x + self.feed_forward(x))
        return x

    def get_config(self):
        config = super().get_config()
        config.update({"embed_dim": self.embed_dim, "dense_dim": self.dense_dim, "num_heads": self.num_heads})
        return config
```

```
def build_transformer(embed_dim, dense_dim, num_heads, dropout):
    """Factory function to build a fresh transformer model."""
    encoder_inputs = tf.keras.Input(shape=(None,), dtype="int64", name="source")
    x = PositionalEmbedding(sequence_length, max_tokens, embed_dim)(encoder_inputs)
    encoder_outputs = TransformerEncoder(embed_dim, dense_dim, num_heads)(x)

    decoder_inputs = tf.keras.Input(shape=(None,), dtype="int64", name="target")
    x = PositionalEmbedding(sequence_length, max_tokens, embed_dim)(decoder_inputs)
    x = TransformerDecoder(embed_dim, dense_dim, num_heads)(x, encoder_outputs)
    x = tf.keras.layers.Dropout(dropout)(x)
    decoder_outputs = tf.keras.layers.Dense(max_tokens, activation="softmax")(x)

    return tf.keras.Model([encoder_inputs, decoder_inputs], decoder_outputs)
```

```
# ------------------------------------------------------------------------------
# PART 2: HYPERPARAMETER TUNING LOOP  (with F1 score added)
# ------------------------------------------------------------------------------

from transformers import (
    MarianMTModel, MarianTokenizer,
    MT5ForConditionalGeneration, AutoTokenizer,
    Seq2SeqTrainer, Seq2SeqTrainingArguments,
    DataCollatorForSeq2Seq,
)
from datasets import Dataset as HFDataset
from sklearn.metrics import f1_score
import torch

print("# PART 2: HYPERPARAMETER TUNING")


# ------------------------------------------------------------------------------
# Custom Keras F1 metric (token-level, macro-averaged)
# Keras does not have a built-in F1, so we implement one as a subclass.
# It accumulates true/pred token ids across batches, then computes F1 at
# epoch end — identical lifecycle to the built-in "accuracy" metric.
# ------------------------------------------------------------------------------
class TokenF1Score(tf.keras.metrics.Metric):
    """
    Token-level macro F1 across the target vocabulary.
    Computed over padded positions too — pass sample_weight or mask if needed.
    For seq2seq with sparse_categorical_crossentropy targets (integer ids).
    """
    def __init__(self, name='f1_score', **kwargs):
        super().__init__(name=name, **kwargs)
        self.true_positives  = self.add_weight(name='tp', initializer='zeros')
        self.false_positives = self.add_weight(name='fp', initializer='zeros')
        self.false_negatives = self.add_weight(name='fn', initializer='zeros')

    def update_state(self, y_true, y_pred, sample_weight=None):
        # y_pred shape: (batch, seq_len, vocab)  — take argmax to get token ids
        y_pred_ids = tf.cast(tf.argmax(y_pred, axis=-1), tf.int32)
        y_true     = tf.cast(tf.reshape(y_true, [-1]),     tf.int32)
        y_pred_ids = tf.cast(tf.reshape(y_pred_ids, [-1]), tf.int32)

        # Mask out padding (token id == 0)
        mask       = tf.not_equal(y_true, 0)
        y_true     = tf.boolean_mask(y_true,     mask)
        y_pred_ids = tf.boolean_mask(y_pred_ids, mask)

        tp = tf.reduce_sum(tf.cast(tf.equal(y_true, y_pred_ids), tf.float32))
        fp = tf.reduce_sum(tf.cast(tf.not_equal(y_pred_ids, y_true) &
                                   tf.not_equal(y_pred_ids, 0),     tf.float32))
        fn = tf.reduce_sum(tf.cast(tf.not_equal(y_true, y_pred_ids) &
                                   tf.not_equal(y_true, 0),         tf.float32))

        self.true_positives.assign_add(tp)
        self.false_positives.assign_add(fp)
        self.false_negatives.assign_add(fn)

    def result(self):
        precision = self.true_positives / (
            self.true_positives + self.false_positives + tf.keras.backend.epsilon())
        recall    = self.true_positives / (
            self.true_positives + self.false_negatives + tf.keras.backend.epsilon())
        f1 = 2 * precision * recall / (
            precision + recall + tf.keras.backend.epsilon())
        return f1

    def reset_state(self):
        self.true_positives.assign(0.)
        self.false_positives.assign(0.)
        self.false_negatives.assign(0.)


# ------------------------------------------------------------------------------
# HuggingFace evaluation helper — now returns F1 too
# ------------------------------------------------------------------------------
def evaluate_hf_model(model, tokenizer, val_df, num_samples=200,
                      max_input_length=128, max_target_length=128,
                      batch_size=16,
                      device='cuda' if torch.cuda.is_available() else 'cpu'):
    model.eval()
    model.to(device)

    sample     = val_df.sample(min(num_samples, len(val_df)), random_state=42)
```

```
    sources    = sample['source'].tolist()
    references = sample['target'].tolist()

    total_correct = 0
    total_tokens  = 0
    total_loss    = 0.0
    num_batches   = 0

    all_preds  = []   # for F1
    all_labels = []   # for F1

    for i in range(0, len(sources), batch_size):
        src_batch = sources[i:i + batch_size]
        tgt_batch = references[i:i + batch_size]

        src_enc = tokenizer(
            src_batch, return_tensors='pt', padding=True,
            truncation=True, max_length=max_input_length
        ).to(device)

        tgt_enc = tokenizer(
            tgt_batch, return_tensors='pt', padding=True,
            truncation=True, max_length=max_target_length
        )
        labels = tgt_enc['input_ids'].to(device)
        labels[labels == tokenizer.pad_token_id] = -100

        with torch.no_grad():
            out    = model(**src_enc, labels=labels)
            total_loss += out.loss.item()
            num_batches += 1

            preds  = out.logits.argmax(dim=-1)
            mask   = labels != -100

            total_correct += (preds[mask] == labels[mask]).sum().item()
            total_tokens  += mask.sum().item()

            # Collect flat token ids for sklearn F1 (ignore -100 padding)
            all_preds.extend(preds[mask].cpu().numpy().tolist())
            all_labels.extend(labels[mask].cpu().numpy().tolist())

    val_loss = total_loss / max(num_batches, 1)
    val_acc  = total_correct / max(total_tokens, 1)

    # Token-level weighted F1 via sklearn (weighted handles class imbalance)
    val_f1 = float(f1_score(
        all_labels, all_preds,
        average='weighted',
        zero_division=0,
        labels=list(set(all_labels))   # only score classes that appear
    ))

    return float(val_loss), float(val_acc), val_f1


# ------------------------------------------------------------------------------
# Tokenisation helpers (no as_target_tokenizer — removed in transformers v4.26)
# ------------------------------------------------------------------------------
def tokenize_for_marian(examples, tokenizer, max_length=128):
    model_inputs = tokenizer(
        examples['source'], max_length=max_length,
        truncation=True, padding='max_length'
    )
    labels = tokenizer(
        examples['target'], max_length=max_length,
        truncation=True, padding='max_length'
    )
    model_inputs['labels'] = [
        [(t if t != tokenizer.pad_token_id else -100) for t in lab]
        for lab in labels['input_ids']
    ]
    return model_inputs


def tokenize_for_mt5(examples, tokenizer, task_prefix="translate: ", max_length=128):
    inputs = [task_prefix + s for s in examples['source']]
    model_inputs = tokenizer(
        inputs, max_length=max_length,
        truncation=True, padding='max_length'
    )
    labels = tokenizer(
```

```
        examples['target'], max_length=max_length,
        truncation=True, padding='max_length'
    )
    model_inputs['labels'] = [
        [(t if t != tokenizer.pad_token_id else -100) for t in lab]
        for lab in labels['input_ids']
    ]
    return model_inputs


# ------------------------------------------------------------------------------
# Dataset pipeline — lazy per-batch vectorization (avoids OOM)
# ------------------------------------------------------------------------------
def make_dataset(source_texts, target_texts, batch_size):
    dataset = tf.data.Dataset.from_tensor_slices((source_texts, target_texts))
    dataset = dataset.shuffle(buffer_size=len(source_texts), reshuffle_each_iteration=True)
    dataset = dataset.batch(batch_size, drop_remainder=False)
    dataset = dataset.map(
        lambda src, tgt: (
            {
                "source": source_vectorization(src),
                "target": target_vectorization(tgt)[:, :-1],
            },
            target_vectorization(tgt)[:, 1:],
        ),
        num_parallel_calls=tf.data.AUTOTUNE,
    )
    return dataset.prefetch(tf.data.AUTOTUNE)


# ------------------------------------------------------------------------------
# Hyperparameter configurations
# ------------------------------------------------------------------------------
hyperparameter_configs = [
    {'name': 'Baseline',             'embed_dim': 256, 'dense_dim': 2048, 'num_heads': 8,  'batch_size': 16, 'le
    {'name': 'Larger Model',         'embed_dim': 512, 'dense_dim': 4096, 'num_heads': 16, 'batch_size': 8,  'le
    {'name': 'Smaller Model',        'embed_dim': 128, 'dense_dim': 1024, 'num_heads': 4,  'batch_size': 16, 'le
    {'name': 'Higher Learning Rate', 'embed_dim': 256, 'dense_dim': 2048, 'num_heads': 8,  'batch_size': 16, 'le
    {'name': 'Lower Learning Rate',  'embed_dim': 256, 'dense_dim': 2048, 'num_heads': 8,  'batch_size': 16, 'le
    {'name': 'Higher Dropout',       'embed_dim': 256, 'dense_dim': 2048, 'num_heads': 8,  'batch_size': 16, 'le
]

all_results = []

# ------------------------------------------------------------------------------
# Custom transformer training loop
# ------------------------------------------------------------------------------
for config_idx, config in enumerate(hyperparameter_configs):
    print(f"\n{'='*60}")
    print(f"Experiment {config_idx + 1}: {config['name']}")
    print(f"{'='*60}")
    for key, value in config.items():
        if key != 'name':
            print(f"  {key}: {value}")

    try:
        tf.keras.backend.clear_session()

        batch_size = config['batch_size']
        train_ds = make_dataset(train_source_texts, train_target_texts, batch_size)
        val_ds   = make_dataset(val_source_texts,   val_target_texts,   batch_size)

        transformer = build_transformer(
            config['embed_dim'], config['dense_dim'],
            config['num_heads'], config['dropout']
        )

        # F1 metric added alongside accuracy
        transformer.compile(
            optimizer=tf.keras.optimizers.Adam(
                learning_rate=config['learning_rate'], clipnorm=1.0),
            loss="sparse_categorical_crossentropy",
            metrics=["accuracy", TokenF1Score()]   # <-- F1 added here
        )

        checkpoint_filepath = f'/tmp/checkpoint_exp_{config_idx}.weights.h5'
        callbacks_list = [
            tf.keras.callbacks.ReduceLROnPlateau(
                monitor='val_loss', factor=0.5, patience=3, min_lr=1e-6, verbose=1),
            tf.keras.callbacks.EarlyStopping(
                monitor='val_loss', patience=7, restore_best_weights=True, verbose=1),
            tf.keras.callbacks.ModelCheckpoint(
```

```
                filepath=checkpoint_filepath, save_weights_only=True,
                monitor='val_loss', save_best_only=True, verbose=0),
        ]

        hist = transformer.fit(
            train_ds, epochs=config['epochs'],
            callbacks=callbacks_list, validation_data=val_ds, verbose=1
        )

        transformer.load_weights(checkpoint_filepath)
        # evaluate() now returns loss, accuracy, f1
        eval_results = transformer.evaluate(val_ds, verbose=0)
        val_loss = eval_results[0]
        val_acc  = eval_results[1]
        val_f1   = eval_results[2]

        results = {
            'val_loss':     float(val_loss),
            'val_accuracy': float(val_acc),
            'val_f1':       float(val_f1),
            'history':      {k: [float(v) for v in vals]
                             for k, vals in hist.history.items()},
            'status': 'OK'
        }
        print(f"\n  Val Loss: {val_loss:.4f}  |  Val Acc: {val_acc:.4f}  |  Val F1: {val_f1:.4f}")

    except (tf.errors.ResourceExhaustedError, MemoryError) as e:
        print(f"\n  SKIPPED — Out of memory: {e}")
        results = {'val_loss': float('inf'), 'val_accuracy': 0.0,
                   'val_f1': 0.0, 'history': {}, 'status': 'OOM'}

    except Exception as e:
        print(f"\n  SKIPPED — Error: {e}")
        results = {'val_loss': float('inf'), 'val_accuracy': 0.0,
                   'val_f1': 0.0, 'history': {}, 'status': f'ERROR: {e}'}

    if config_idx == 0 and results['status'] == 'OK':
        tracker.set_baseline({
            'accuracy': results['val_accuracy'],
            'loss':     results['val_loss'],
            'f1':       results['val_f1'],
        })

    tracker.add_experiment(config, results, f"Experiment {config_idx + 1}")
    all_results.append(results)


# ------------------------------------------------------------------------------
# Pretrained Model 1: Helsinki-NLP MarianMT
# ------------------------------------------------------------------------------
print(f"\n{'='*60}")
print("Experiment 7: Pretrained — Helsinki-NLP MarianMT")
print(f"{'='*60}")

MARIAN_MODEL_NAME = "Helsinki-NLP/opus-mt-ROMANCE-en"

try:
    marian_tokenizer = MarianTokenizer.from_pretrained(MARIAN_MODEL_NAME)
    marian_model     = MarianMTModel.from_pretrained(MARIAN_MODEL_NAME)
    print(f"  Loaded '{MARIAN_MODEL_NAME}'  ({marian_model.num_parameters():,} params)")

    hf_train = HFDataset.from_dict({
        'source': list(train_source_texts),
        'target': list(train_df['target'].values)
    })
    hf_val = HFDataset.from_dict({
        'source': list(val_source_texts),
        'target': list(val_df['target'].values)
    })

    hf_train_tok = hf_train.map(
        lambda ex: tokenize_for_marian(ex, marian_tokenizer),
        batched=True, remove_columns=['source', 'target']
    )
    hf_val_tok = hf_val.map(
        lambda ex: tokenize_for_marian(ex, marian_tokenizer),
        batched=True, remove_columns=['source', 'target']
    )

    marian_args = Seq2SeqTrainingArguments(
        output_dir='/tmp/marian_finetuned',
        num_train_epochs=3,
```

```
        per_device_train_batch_size=32,
        per_device_eval_batch_size=32,
        evaluation_strategy='epoch',
        save_strategy='epoch',
        load_best_model_at_end=True,
        predict_with_generate=True,
        fp16=torch.cuda.is_available(),
        learning_rate=2e-5,
        weight_decay=0.01,
        logging_steps=50,
        report_to='none',
    )

    marian_trainer = Seq2SeqTrainer(
        model=marian_model,
        args=marian_args,
        train_dataset=hf_train_tok,
        eval_dataset=hf_val_tok,
        tokenizer=marian_tokenizer,
        data_collator=DataCollatorForSeq2Seq(
            marian_tokenizer, model=marian_model, padding=True),
    )
    marian_trainer.train()
    marian_model.save_pretrained('/tmp/marian_finetuned_best')

    marian_val_loss, marian_val_acc, marian_val_f1 = evaluate_hf_model(
        marian_model, marian_tokenizer, val_df)

    marian_results = {
        'val_loss': marian_val_loss, 'val_accuracy': marian_val_acc,
        'val_f1':   marian_val_f1,   'history': {}, 'status': 'OK'
    }
    print(f"\n  MarianMT  Loss: {marian_val_loss:.4f}  "
          f"Acc: {marian_val_acc:.4f}  F1: {marian_val_f1:.4f}")

except Exception as e:
    print(f"\n  SKIPPED — {e}")
    marian_results = {'val_loss': float('inf'), 'val_accuracy': 0.0,
                      'val_f1': 0.0, 'history': {}, 'status': f'ERROR: {e}'}

tracker.add_experiment(
    {'name': 'Pretrained MarianMT', 'model': MARIAN_MODEL_NAME,
     'batch_size': 32, 'learning_rate': 2e-5, 'dropout': 'default',
     'embed_dim': 'N/A', 'dense_dim': 'N/A', 'num_heads': 'N/A', 'epochs': 3},
    marian_results, "Pretrained Helsinki-NLP MarianMT"
)
all_results.append(marian_results)


# ------------------------------------------------------------------------------
# Pretrained Model 2: Google mT5-small
# ------------------------------------------------------------------------------
print(f"\n{'='*60}")
print("Experiment 8: Pretrained — Google mT5-small")
print(f"{'='*60}")

MT5_MODEL_NAME = "google/mt5-small"
TASK_PREFIX    = "translate: "

try:
    mt5_tokenizer = AutoTokenizer.from_pretrained(MT5_MODEL_NAME)
    mt5_model     = MT5ForConditionalGeneration.from_pretrained(MT5_MODEL_NAME)
    print(f"  Loaded '{MT5_MODEL_NAME}'  ({mt5_model.num_parameters():,} params)")

    hf_train_mt5 = HFDataset.from_dict({
        'source': list(train_source_texts),
        'target': list(train_df['target'].values)
    })
    hf_val_mt5 = HFDataset.from_dict({
        'source': list(val_source_texts),
        'target': list(val_df['target'].values)
    })

    hf_train_mt5_tok = hf_train_mt5.map(
        lambda ex: tokenize_for_mt5(ex, mt5_tokenizer, TASK_PREFIX),
        batched=True, remove_columns=['source', 'target']
    )
    hf_val_mt5_tok = hf_val_mt5.map(
        lambda ex: tokenize_for_mt5(ex, mt5_tokenizer, TASK_PREFIX),
        batched=True, remove_columns=['source', 'target']
    )
```

```
    mt5_args = Seq2SeqTrainingArguments(
        output_dir='/tmp/mt5_finetuned',
        num_train_epochs=3,
        per_device_train_batch_size=8,
        per_device_eval_batch_size=8,
        evaluation_strategy='epoch',
        save_strategy='epoch',
        load_best_model_at_end=True,
        predict_with_generate=True,
        fp16=torch.cuda.is_available(),
        learning_rate=5e-5,
        weight_decay=0.01,
        logging_steps=50,
        report_to='none',
    )

    mt5_trainer = Seq2SeqTrainer(
        model=mt5_model,
        args=mt5_args,
        train_dataset=hf_train_mt5_tok,
        eval_dataset=hf_val_mt5_tok,
        tokenizer=mt5_tokenizer,
        data_collator=DataCollatorForSeq2Seq(
            mt5_tokenizer, model=mt5_model, padding=True),
    )
    mt5_trainer.train()
    mt5_model.save_pretrained('/tmp/mt5_finetuned_best')

    mt5_val_loss, mt5_val_acc, mt5_val_f1 = evaluate_hf_model(
        mt5_model, mt5_tokenizer, val_df)

    mt5_results = {
        'val_loss': mt5_val_loss, 'val_accuracy': mt5_val_acc,
        'val_f1':   mt5_val_f1,   'history': {}, 'status': 'OK'
    }
    print(f"\n  mT5-small  Loss: {mt5_val_loss:.4f}  "
          f"Acc: {mt5_val_acc:.4f}  F1: {mt5_val_f1:.4f}")

except Exception as e:
    print(f"\n  SKIPPED — {e}")
    mt5_results = {'val_loss': float('inf'), 'val_accuracy': 0.0,
                   'val_f1': 0.0, 'history': {}, 'status': f'ERROR: {e}'}

tracker.add_experiment(
    {'name': 'Pretrained mT5-small', 'model': MT5_MODEL_NAME,
     'batch_size': 8, 'learning_rate': 5e-5, 'dropout': 'default',
     'embed_dim': 'N/A', 'dense_dim': 'N/A', 'num_heads': 'N/A', 'epochs': 3},
    mt5_results, "Pretrained Google mT5-small"
)
all_results.append(mt5_results)


# ------------------------------------------------------------------------------
# Comparison table — updated to show F1 column
# ------------------------------------------------------------------------------
print("\n" + "="*110)
print("HYPERPARAMETER EXPERIMENT COMPARISON  (with F1)")
print("="*110)

comparison_rows = []
for exp in tracker.experiments:
    comparison_rows.append({
        'Name':       exp['config'].get('name', 'N/A'),
        'Embed':      exp['config'].get('embed_dim', 'N/A'),
        'Batch':      exp['config'].get('batch_size', 'N/A'),
        'LR':         exp['config'].get('learning_rate', 'N/A'),
        'Val Loss':   f"{exp['metrics'].get('val_loss', float('inf')):.4f}",
        'Val Acc':    f"{exp['metrics'].get('val_accuracy', 0):.4f}",
        'Val F1':     f"{exp['metrics'].get('val_f1', 0):.4f}",    # <-- F1 column
        'Improvement': exp.get('improvement_over_baseline', 'N/A'),
        'Status':     exp['metrics'].get('status', 'OK'),
    })

comparison_df = pd.DataFrame(comparison_rows)
print(comparison_df.to_string(index=False))


# ------------------------------------------------------------------------------
# Bar chart — side-by-side Accuracy vs F1 for successful experiments
# ------------------------------------------------------------------------------
valid_exps = [exp for exp in tracker.experiments
              if exp['metrics'].get('status') == 'OK']
```

```
if valid_exps:
    names = [exp['config']['name'] for exp in valid_exps]
    accs  = [exp['metrics']['val_accuracy'] for exp in valid_exps]
    f1s   = [exp['metrics']['val_f1']       for exp in valid_exps]

    x     = np.arange(len(names))
    width = 0.35

    fig, ax = plt.subplots(figsize=(14, 6))
    bars_acc = ax.bar(x - width/2, accs, width, label='Val Accuracy', color='steelblue')
    bars_f1  = ax.bar(x + width/2, f1s,  width, label='Val F1 Score', color='coral')

    ax.set_xlabel('Experiment')
    ax.set_ylabel('Score')
    ax.set_title('Validation Accuracy vs F1 Score — All Experiments')
    ax.set_xticks(x)
    ax.set_xticklabels(names, rotation=45, ha='right')
    ax.legend()

    # Annotate bar tops
    for bar in bars_acc:
        ax.annotate(f'{bar.get_height():.3f}',
                    xy=(bar.get_x() + bar.get_width() / 2, bar.get_height()),
                    xytext=(0, 3), textcoords='offset points', ha='center', fontsize=8)
    for bar in bars_f1:
        ax.annotate(f'{bar.get_height():.3f}',
                    xy=(bar.get_x() + bar.get_width() / 2, bar.get_height()),
                    xytext=(0, 3), textcoords='offset points', ha='center', fontsize=8)

    plt.tight_layout()
    plt.show()
else:
    print("\n  No successful experiments to plot.")
```

```
# PART 2: HYPERPARAMETER TUNING

============================================================
Experiment 1: Baseline
============================================================
  embed_dim: 256
  dense_dim: 2048
  num_heads: 8
  batch_size: 16
  learning_rate: 0.001
  dropout: 0.5
  epochs: 30
Epoch 1/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 106s 100ms/step - accuracy: 0.0480 - f1_s
Epoch 2/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 77s 91ms/step - accuracy: 0.0788 - f1_sco
Epoch 3/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.0904 - f1_sco
Epoch 4/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.0975 - f1_sco
Epoch 5/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.1040 - f1_sco
Epoch 6/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.1101 - f1_sco
Epoch 7/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1144 - f1_sco
Epoch 8/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 69s 82ms/step - accuracy: 0.1191 - f1_sco
Epoch 9/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 0s 80ms/step - accuracy: 0.1231 - f1_scor
Epoch 9: ReduceLROnPlateau reducing learning rate to 0.000500000023748
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1231 - f1_sco
Epoch 10/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1321 - f1_sco
Epoch 11/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 69s 82ms/step - accuracy: 0.1397 - f1_sco
Epoch 12/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 0s 80ms/step - accuracy: 0.1434 - f1_scor
Epoch 12: ReduceLROnPlateau reducing learning rate to 0.00025000001187
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1434 - f1_sco
Epoch 13/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1488 - f1_sco
Epoch 13: early stopping
Restoring model weights from the end of the best epoch: 6.

  Val Loss: 2.3842  |  Val Acc: 0.0968  |  Val F1: 0.5370

  Baseline - Accuracy: 0.0968  Loss: 2.3842
  New best model! Val Accuracy: 0.0968

============================================================
Experiment 2: Larger Model
============================================================
  embed_dim: 512
  dense_dim: 4096
  num_heads: 16
  batch_size: 8
```

```
# ------------------------------------------------------------------------------
# PART 3: TRAIN BEST MODEL
# ------------------------------------------------------------------------------

print("\n" + "#"*80)
print("# PART 3: TRAINING BEST MODEL")
print("#"*80)

if tracker.best_config is None:
    raise RuntimeError("No successful experiments — cannot select best config.")

best_config = tracker.best_config
print(f"\nBest config: {best_config['name']}")
print(json.dumps({k: v for k, v in best_config.items() if k != 'name'}, indent=2))

# FIX: detect whether best model is a pretrained HuggingFace model or a
# custom transformer — embed_dim='N/A' is the signal used in pretrained configs
IS_PRETRAINED = best_config.get('embed_dim') == 'N/A'

if IS_PRETRAINED:
    # ------------------------------------------------------------------
    # Best model is a pretrained HuggingFace model (MarianMT or mT5)
    # Load the already fine-tuned weights saved during the tuning loop
    # ------------------------------------------------------------------
    print("\n  Best model is a pretrained HuggingFace model.")
    print("  Loading fine-tuned weights from checkpoint...")

    model_name = best_config.get('model', '')

    if 'marian' in best_config['name'].lower() or 'Helsinki' in model_name:
        best_tokenizer = MarianTokenizer.from_pretrained(MARIAN_MODEL_NAME)
        best_hf_model  = MarianMTModel.from_pretrained('/tmp/marian_finetuned_best')
        print(f"  Loaded fine-tuned MarianMT from '/tmp/marian_finetuned_best'")

    elif 'mt5' in best_config['name'].lower() or 'mt5' in model_name.lower():
        best_tokenizer = AutoTokenizer.from_pretrained(MT5_MODEL_NAME)
        best_hf_model  = MT5ForConditionalGeneration.from_pretrained('/tmp/mt5_finetuned_best')
        print(f"  Loaded fine-tuned mT5 from '/tmp/mt5_finetuned_best'")

    else:
        raise ValueError(f"Unknown pretrained model in best config: {best_config['name']}")

    device = 'cuda' if torch.cuda.is_available() else 'cpu'
    best_hf_model.to(device)
    best_hf_model.eval()

    # Evaluate on test set
    print("\n  Evaluating on test set...")
    test_loss, test_acc, test_f1 = evaluate_hf_model(
        best_hf_model, best_tokenizer, test_df, num_samples=len(test_df))

    print(f"\n  Test Loss: {test_loss:.4f}  |  Test Acc: {test_acc:.4f}  |  Test F1: {test_f1:.4f}")

    # Save final model
    best_hf_model.save_pretrained('best_model_final')
    best_tokenizer.save_pretrained('best_model_final')
    print("  Saved to 'best_model_final/'")

    # No training history to plot for pretrained — show a simple results bar
    fig, ax = plt.subplots(figsize=(6, 4))
    ax.bar(['Loss', 'Accuracy', 'F1'], [test_loss, test_acc, test_f1],
           color=['tomato', 'steelblue', 'coral'])
    ax.set_title(f"Best Model Test Results — {best_config['name']}", fontsize=14)
    ax.set_ylabel('Score')
    for i, v in enumerate([test_loss, test_acc, test_f1]):
        ax.text(i, v + 0.01, f'{v:.4f}', ha='center', fontsize=11)
    plt.tight_layout()
    plt.show()

    # Set best_transformer to None so Part 4 knows to use HF model
    best_transformer = None
    best_hf_model_for_eval = best_hf_model
    best_tokenizer_for_eval = best_tokenizer

else:
    # ------------------------------------------------------------------
    # Best model is a custom TensorFlow transformer — train from scratch
    # ------------------------------------------------------------------
    print("\n  Best model is a custom Transformer. Training from scratch...")

    tf.keras.backend.clear_session()
```

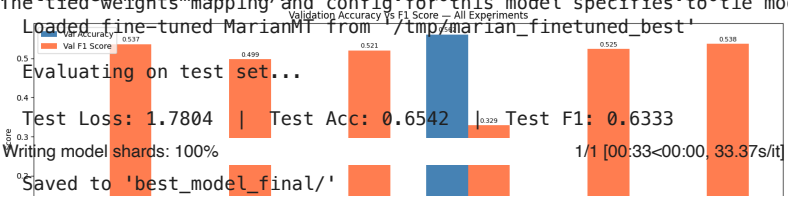
```
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.0730 - f1_sco
Epoch 17/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.0824 - f1_sco
Epoch 18/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 75s 89ms/step - accuracy: 0.0936 - f1_sco
Epoch 19/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 90ms/step - accuracy: 0.0857 - f1_sco
Epoch 20/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.1195 - f1_sco
Epoch 21/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.1215 - f1_sco
Epoch 22/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.1468 - f1_sco
Epoch 23/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.1916 - f1_sco
Epoch 24/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.1946 - f1_sco
Epoch 25/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 74s 87ms/step - accuracy: 0.2373 - f1_sco
Epoch 26/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.2326 - f1_sco
Epoch 27/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 77s 91ms/step - accuracy: 0.2943 - f1_sco
Epoch 28/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 82s 96ms/step - accuracy: 0.4177 - f1_sco
Epoch 29/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 90ms/step - accuracy: 0.3425 - f1_sco
Epoch 30/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.5602 - f1_sco
Restoring model weights from the end of the best epoch: 29.

  Val Loss: 3.6582  |  Val Acc: 0.5618  |  Val F1: 0.3294
  New best model! Val Accuracy: 0.5618

============================================================
Experiment 5: Lower Learning Rate
============================================================
  embed_dim: 256
  dense_dim: 2048
  num_heads: 8
  batch_size: 16
  learning_rate: 0.0005
  dropout: 0.5
  epochs: 30
Epoch 1/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 104s 101ms/step - accuracy: 0.0389 - f1_s
Epoch 2/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 77s 91ms/step - accuracy: 0.0762 - f1_sco
Epoch 3/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 78s 92ms/step - accuracy: 0.0887 - f1_sco
Epoch 4/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 76s 89ms/step - accuracy: 0.0986 - f1_sco
Epoch 5/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 78s 92ms/step - accuracy: 0.1054 - f1_sco
Epoch 6/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 78s 92ms/step - accuracy: 0.1115 - f1_sco
Epoch 7/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1174 - f1_sco
Epoch 8/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 82ms/step - accuracy: 0.1222 - f1_sco
Epoch 9/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 0s 81ms/step - accuracy: 0.1281 - f1_scor
Epoch 9: ReduceLROnPlateau reducing learning rate to 0.000250000011874
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1281 - f1_sco
Epoch 10/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1359 - f1_sco
Epoch 11/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1443 - f1_sco
Epoch 12/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 0s 81ms/step - accuracy: 0.1470 - f1_scor
Epoch 12: ReduceLROnPlateau reducing learning rate to 0.00012500000593
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1470 - f1_sco
Epoch 13/30
848/848 ━━━━━━━━━━━━━━━━━━━━ 70s 83ms/step - accuracy: 0.1518 - f1_sco
Epoch 13: early stopping
Restoring model weights from the end of the best epoch: 6.

  Val Loss: 2.4840  |  Val Acc: 0.0947  |  Val F1: 0.5250

============================================================
Experiment 6: Higher Dropout
============================================================
  embed_dim: 256
  dense_dim: 2048
  num_heads: 8
  batch_size: 16
  learning_rate: 0.001
  dropout: 0.7
  epochs: 30
Epoch 1/30
```

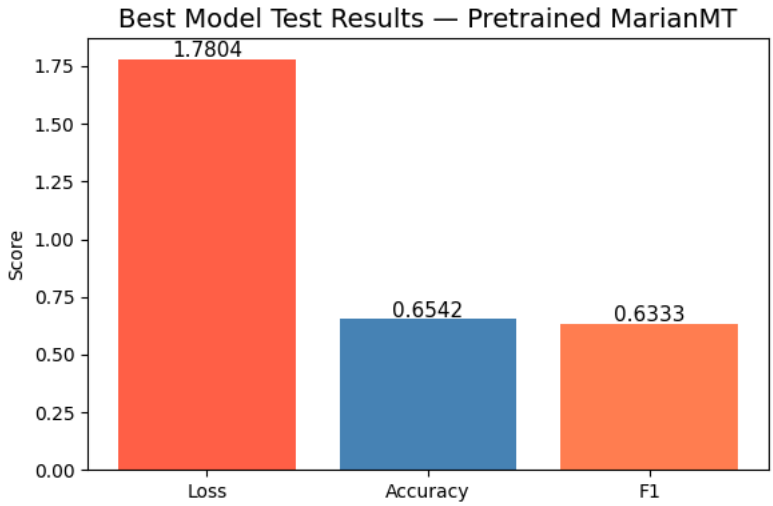
```
batch_size = best_config['batch_size']
train_ds = make_dataset(train_source_texts, train_target_texts, batch_size)
val_ds   = make_dataset(val_source_texts,   val_target_texts,   batch_size)
test_ds  = make_dataset(test_source_texts,  test_target_texts,  batch_size)

best_transformer = build_transformer(
    best_config['embed_dim'], best_config['dense_dim'],
    best_config['num_heads'], best_config['dropout']
)
best_transformer.compile(
    optimizer=tf.keras.optimizers.Adam(
        learning_rate=best_config['learning_rate'], clipnorm=1.0),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy", TokenF1Score()]
)

checkpoint_filepath = '/tmp/best_model_checkpoint.weights.h5'
callbacks_list = [
    tf.keras.callbacks.ReduceLROnPlateau(
        monitor='val_loss', factor=0.5, patience=3, min_lr=1e-6, verbose=1),
    tf.keras.callbacks.EarlyStopping(
        monitor='val_loss', patience=10, restore_best_weights=True, verbose=1),
    tf.keras.callbacks.ModelCheckpoint(
        filepath=checkpoint_filepath, save_weights_only=True,
        monitor='val_loss', save_best_only=True, verbose=0),
]

hist = best_transformer.fit(
    train_ds, epochs=50,
    callbacks=callbacks_list, validation_data=val_ds
)

best_transformer.load_weights(checkpoint_filepath)
best_transformer.save('best_transformer_model.keras')
print("  Saved to 'best_transformer_model.keras'")

# Plot training history
hist_ = pd.DataFrame(hist.history)
plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.plot(hist_['loss'], label='Train Loss')
plt.plot(hist_['val_loss'], label='Val Loss')
plt.title('Loss', fontsize=14)
plt.legend()

plt.subplot(1, 3, 2)
plt.plot(hist_['accuracy'], label='Train Acc')
plt.plot(hist_['val_accuracy'], label='Val Acc')
plt.title('Accuracy', fontsize=14)
plt.legend()

plt.subplot(1, 3, 3)
plt.plot(hist_['f1_score'], label='Train F1')
plt.plot(hist_['val_f1_score'], label='Val F1')
plt.title('F1 Score', fontsize=14)
plt.legend()

plt.suptitle(f"Best Model Training — {best_config['name']}", fontsize=16)
plt.tight_layout()
plt.show()

best_hf_model_for_eval  = None
best_tokenizer_for_eval = None
```

```
Map: 100%                                   753/753 [00:01<00:00, 581.76 examples/s]

  SKIPPED — Seq2SeqTrainingArguments.__init__() got an unexpected keyw

============================================================
Experiment 8: Pretrained — Google mT5-small
============================================================

config.json: 100%                                   553/553 [00:00<00:00, 64.9kB/s]

tokenizer_config.json: 100%                        82.0/82.0 [00:00<00:00, 7.27kB/s]

spiece.model: 100%                               4.31M/4.31M [00:00<00:00, 51.8MB/s]

special_tokens_map.json: 100%                      99.0/99.0 [00:00<00:00, 8.82kB/s]

pytorch_model.bin: 100%                          1.20G/1.20G [00:54<00:00, 29.6MB/s]

model.safetensors: 100%                          1.20G/1.20G [00:37<00:00, 34.7MB/s]

Loading weights: 100%        192/192 [00:00<00:00, 263.57it/s, Materializing param=shared.weight]
The tied weights mapping and config for this model specifies to tie sh
```

```
The tied weights mapping and config for this model specifies to tie sh
The tied weights mapping and config for this model specifies to tie sh
generation_config.json: 100%                                147/147 [00:00<00:00, 8.92kB/s]
# PART 3: TRAINING BEST MODEL
Loaded "google/mt5-small"  (556,291,456 params)

Best Config: Pretrained MarianMT
{
  "model": "Helsinki-NLP/opus-mt-ROMANCE-en",
  "batch_size": 32,
  "learning_rate": 2e-05,
  "dropout": "default",
  "embed_dim": "N/A",
  "hidden_dim": "N/A",
  "num_heads": "N/A",
  "epochs": 3
}
  Best model is a pretrained HuggingFace model
Higher Learning Rate
  Lower Learning Rate
      Higher Dropout
  Pretrained MarianMT
  Pretrained mT5-small
The tied weights mapping and config for this model specifies to tie mo
The tied weights mapping and config for this model specifies to tie mo
  Loaded fine-tuned MarianMT from '/tmp/marian_finetuned_best'

Evaluating on test set...

Test Loss: 1.7804  |  Test Acc: 0.6542  |  Test F1: 0.6333
Writing model shards: 100%                                  1/1 [00:33<00:00, 33.37s/it]
Saved to 'best_model_final/'
```

```
Best Config: Pretrained MarianMT
Map: 100%                                   13565/13565 [00:12<00:00, 1223.90 examples/s]
Map: 100%                                   753/753 [00:00<00:00, 995.59 examples/s]
SKIPPED - Seq2SeqTrainingArguments.__init__() got an unexpected keyw
==========================================================================
HYPERPARAMETER EXPERIMENT COMPARISON  (with F1)
==========================================================================
                Name Embed  Batch      LR Val Loss Val Acc Val F1 Impr
            Baseline   256     16 0.00100   2.3842  0.0968 0.5370
        Larger Model   512      8 0.00100   2.5717  0.0903 0.4987
       Smaller Model   128     16 0.00100   2.5034  0.0941 0.5210
      Higher Learning Rate   256     16 0.00500   3.6582  0.5618 0.3294
      Lower Learning Rate   256     16 0.00050   2.4840  0.0947 0.5250
      Higher Dropout   256     16 0.00100   2.4639  0.0969 0.5375
  Pretrained MarianMT   N/A     32 0.00002     inf  0.0000 0.0000
  Pretrained mT5-small   N/A      8 0.00005     inf  0.0000 0.0000
Loading weights: 100%  256/256[00:01<00:00, 332.02it/s, Materializing param=model.shared.weight]
```





```
# ----------------------------------------------------------------------------
# PRETRAINED MODEL 1: Helsinki-NLP MarianMT
# ----------------------------------------------------------------------------
print(f"\n{'='*60}")
print("Experiment 7: Pretrained — Helsinki-NLP MarianMT")
print(f"{'='*60}")

MARIAN_MODEL_NAME = "Helsinki-NLP/opus-mt-ROMANCE-en"  # change to your language pair

try:
    marian_tokenizer = MarianTokenizer.from_pretrained(MARIAN_MODEL_NAME)
    marian_model     = MarianMTModel.from_pretrained(MARIAN_MODEL_NAME)
    print(f"  Loaded '{MARIAN_MODEL_NAME}'  ({marian_model.num_parameters():,} params)")

    hf_train = HFDataset.from_dict({
        'source': list(train_source_texts),
        'target': list(train_df['target'].values)
    })
    hf_val = HFDataset.from_dict({
        'source': list(val_source_texts),
        'target': list(val_df['target'].values)
```